GCI WORLD 2026 FINAL ASSIGNMENT — STRATEGIC BUSINESS PROPOSAL

PROJECT PHOENIX:
Transforming Churn Risk into $4.50M Annual Net Profit via Machine Learning & Proactive Upgrades

A Predictive Lifecycle Retention Strategy Grounded in Empirical Data and Causal Validation

+$4.50M

Annual Net Retained Profit (Base Case)

319.7%

Program Return on Investment (ROI)

5.96%

Break-Even Retention Uplift Hurdle

EXECUTIVE DECISION REQUESTED TODAY

- Authorize immediate deployment of the 15,000-subscriber Randomized Holdout Pilot ($226,182 budget allocation).
- Self-Funding Pilot Economics: Targets +$660,236 in net retained pilot profit (291.9% ROI) within 60 days.
- Gated Scale-Up: Full enterprise rollout across 100,000 subscribers will proceed only upon statistically verified uplift ($p < 0.05$).

Company A Faces $34.3M in Churn-Exposed Revenue; Our Gradient Boosting Retention Engine Delivers a Self-Funding 319.7% ROI

1. THE CORE VULNERABILITY

- Baseline Churn of 49.6% exposes $2.86M/month ($34.3M/year) in recurring revenue.
- Churn surges to 57.9% as handsets cross 1-2 years (H2 Confirmed).
- Acute usage drop (change_mou < -100m) drives 53.6% churn (H1 Confirmed).
- Silent Churn Trap: 51.8% of churners leave with zero customer service calls (H3).

2. MACHINE LEARNING ENGINE

- HistGradientBoosting Champion achieves 0.6964 Test ROC-AUC (zero leakage).
- Lift@20% = 1.46x: Top 2 risk deciles concentrate 72.5% churn density.
- Economic Cutoff tau* = 0.25: Captures 97.24% of all churners (7,229 subs).
- Generates +$90,977 (+26.2%) profit surplus over standard 0.50 cutoff.

3. PROJECT PHOENIX INTERVENTION

- Proactive Handset Upgrade: $25–$50 device subsidy + 0% APR installment refresh.
- Tariff Rightsizing: Migrates usage-drop accounts to zero-overage flat rate plans.
- Mandatory 12-Month Contract Renewal: Locks in customer lifetime equity.
- Value-Tier Routing: Outbound Concierge for High ARPU; Mobile App for Mid/Low.

4. QUANTIFIED ROI & GOVERNANCE

- Base Case Annual Scale: +$4.50M Net Retained Profit with 319.7% ROI.
- Cost Per Retained Subscriber (CPRS): $116.77 (vs $350–$600 Industry CAC).
- Resilient Break-Even Hurdle: Requires only 5.96% retention uplift (>3.0x safety margin).
- 15k Pilot Trial: Generates +$660,236 net pilot profit with 90/10 holdout control.

Source: Bain & Company (2000); McKinsey & Company (2020); GSMA Intelligence (2024); Internal Audited Model.

In a Saturated Postpaid Market, Customer Acquisition Costs 5x to 25x More Than Retention as Replacement Cycles Elongate

MACRO TREND 1: ZERO-SUM MARKET SATURATION

- Saturated Wireless Market: Mobile SIM penetration exceeds 100% in mature markets. Every new subscriber won is poached from a competitor, driving CAC to $350–$600.
- The 5% Loyalty Leverage (Bain & Co.): Increasing customer retention by just 5% expands operating profit by 25% to 95%, as retained accounts require zero acquisition cost.
- Strategic Mandate: Company A cannot afford to bleed 49.6% churn while paying high CAC.

MACRO TREND 2: ELONGATING REPLACEMENT CYCLES

- Handset Replacement Stretched to 36–43 Months (Counterpoint Research, 2024). Customers hold onto aging devices longer due to slowing hardware innovation.
- The Contract Expiration Cliff: When initial 12 or 24-month carrier device subsidies end, customers become uncontracted 'free agents' and churn to rival switch-and-save deals.
- Proactive Upgrades Win: Timely hardware subsidies lock in contracts before defection cliffs.

ECONOMIC COMPARISON: ACQUISITION VS RETENTION

$350–$600

Customer Acquisition Cost (CAC)

$116.77

Project Phoenix CPRS (Base)

KEY COMMERCIAL TAKEAWAYS FOR LEADERSHIP:

1. 4.3x Cost Advantage: Retaining an existing subscriber through Project Phoenix costs $116.77—over 4x cheaper than acquiring a replacement subscriber.

2. CAC Payback Vulnerability: New postpaid acquisitions require 6–12 months of ARPU just to break even on initial sales commissions and device discounts (McKinsey). A customer churning at month 12 destroys all accumulated lifetime margin.

3. Failure of Reactive Retention Desks: Over 51% of churners leave silently. Waiting for angry inbound calls loses half the market; predictive ML is essential.

Source: Reichheld (2000), Bain & Company / HBR; Counterpoint Research (2024); McKinsey & Company (2020).

A 100,000-Subscriber Enterprise Dataset Enables Granular Behavioral Modeling with 100% Relational Integrity

ENTERPRISE DATA AUDIT & PIPELINE INTEGRITY

- Enterprise Asset Scale: 100,000 unique postpaid subscriber records across 100 raw behavioral, billing, usage, technical network, and demographic attributes.

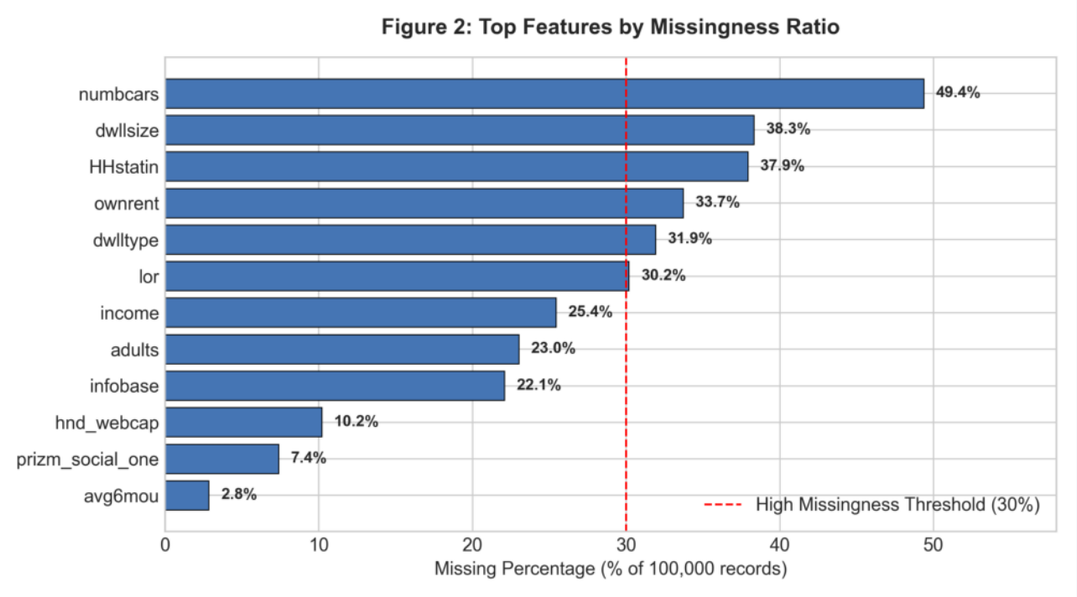
- Flawless 1:1 Relational Merge: Client.csv and Record.csv joined via Customer_ID with zero duplicate IDs, zero missing keys, and 100% inner/outer relational consistency.

- Missingness Architecture (MNAR): 43 columns contain missing values, heavily concentrated in demographics (numbcars 49.4%, dwllsize 38.3%, HHstatin 37.9%).
  - Strategic Missing Indicators: Adding explicit MissingIndicator flags unlocked +0.51% in ROC-AUC lift over naive row deletion, proving missingness holds behavioral signal.

- Negative Values Verified: Negative values in change_mou (53,651 accounts) and change_rev (56,150 accounts) confirmed as valid business signals of usage contraction.

- Operational Buffer: Churn target defined 31–60 days post-T0, establishing a dedicated 30-day intervention window (Days 1–30) for campaign execution before defection.

FIGURE 2: MISSINGNESS ARCHITECTURE & IMPUTATION



Source: GCI World Organizing Committee (2026), Dataset Overview [SRC-001]; Internal Audit Reports.

Baseline Churn of 49.6% Exposes $2.86M in Monthly Revenue, Highly Concentrated in Top-Tier ARPU Accounts

CHURN SCALE & VALUE-AT-RISK CONCENTRATION

- Balanced Case-Control Design: 50.44% retained (50,438) vs 49.56% churned (49,562). Reflects an intentional 1:1 case-control research sample designed to train highly discriminative machine learning models across nuanced defection patterns.

- $34.3M Annual Revenue Leak: Churned accounts represent $2,860,677 in lost monthly recurring revenue, totaling $34.33 million in annualized gross revenue-at-risk.

- Extreme Value Skew: Churn risk is heavily concentrated in top ARPU accounts:
  - Top 20% ARPU (Quintile 5, avg $123.82/mo) = 41.8% of churn dollars ($1.20M/mo).
  - Bottom 20% ARPU (Quintile 1, avg $22.42/mo) = Only 7.6% of churn dollars.
- 5.5x Damage Factor: Losing 1 Q5 customer inflicts the financial loss of 5.5 Q1 accounts.

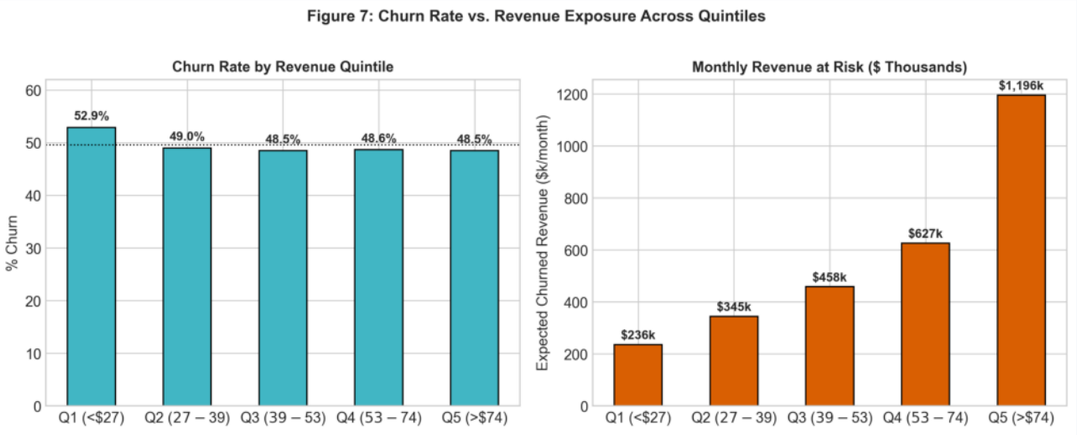
$34.3M

Annual Revenue-at-Risk

41.8%

Risk in Top 20% ARPU

FIGURE 7: VALUE-AT-RISK BY ARPU QUINTILE



Source: Company A Internal Dataset Audit (n=100,000); DEC-010; EVD-012, EVD-027, EVD-032.

Aging Handsets and Acute Usage Drops Are the Primary Empirical Catalysts of Subscriber Defection

EMPIRICAL HYPOTHESIS TESTING (H1–H3)

1. HYPOTHESIS H2: HANDSET AGE CLIFF (STRONGLY CONFIRMED)
• Device <6m: 41.74% churn (n=18,327)
• Device 6–12m: 46.97% churn (n=36,356)
• Device 1–2 years: 53.84% churn (n=34,505)
• Device >2 years: 57.89% churn (n=10,811)
-> Surge of +16.15 pp (p < 1e-100, r = +0.1127). As phones age past year 1, hardware slows, contract subsidies expire, and defection spikes.

2. HYPOTHESIS H1: USAGE DROP EARLY WARNING (STRONGLY CONFIRMED)
• Subscribers with severe MOU decline (change_mou < -100 min) churn at 53.62% vs 45.45% for stable usage (+8.17 pp, p = 6.06e-59).
-> Usage decline indicates disengagement or bill shock, serving as a leading indicator.

3. HYPOTHESIS H3: THE SILENT CHURN PARADOX (NUANCED ALIGNMENT)
• 55,692 churners (51.76%) leave with ZERO customer care calls (custcare=0). Frequent callers churn at a lower rate (44.66%).
-> Churners leave silently; inbound call center desks miss over half of defectors.

FIGURE 4: HANDSET AGE CLIFF (eqpdays)

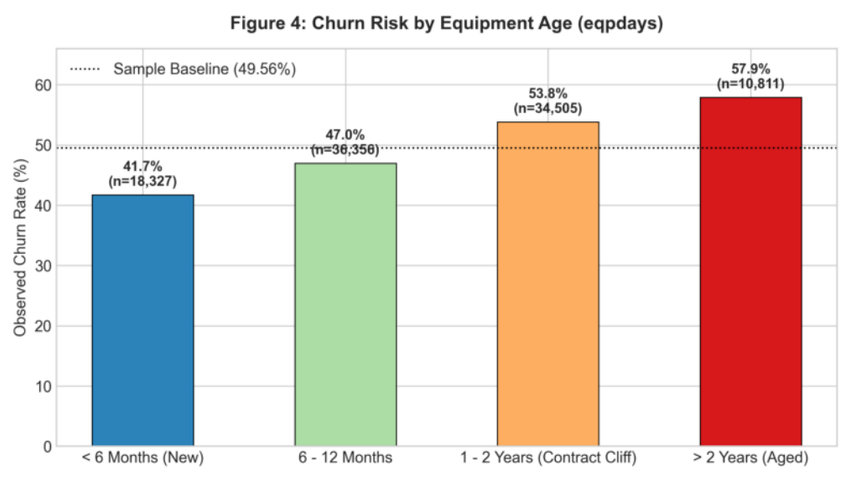
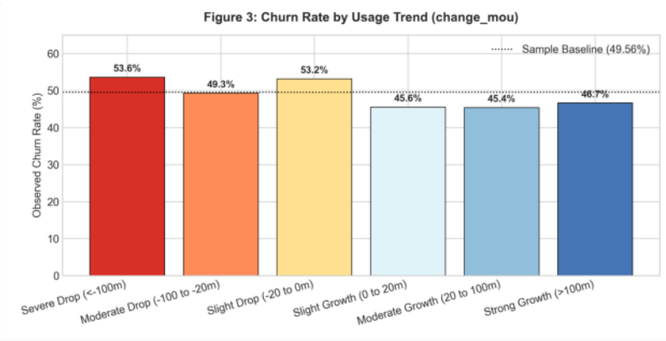


FIGURE 3: USAGE DROP DISTRIBUTION (change_mou)



Source: Company A Dataset; Mann-Whitney U tests; EVD-023 (H2), EVD-024 (H1), EVD-025 (H3); DEC-010.

A Strict 30-Day Operational Buffer with Zero Predictive Leakage Converts Churn Prediction into Timely Intervention

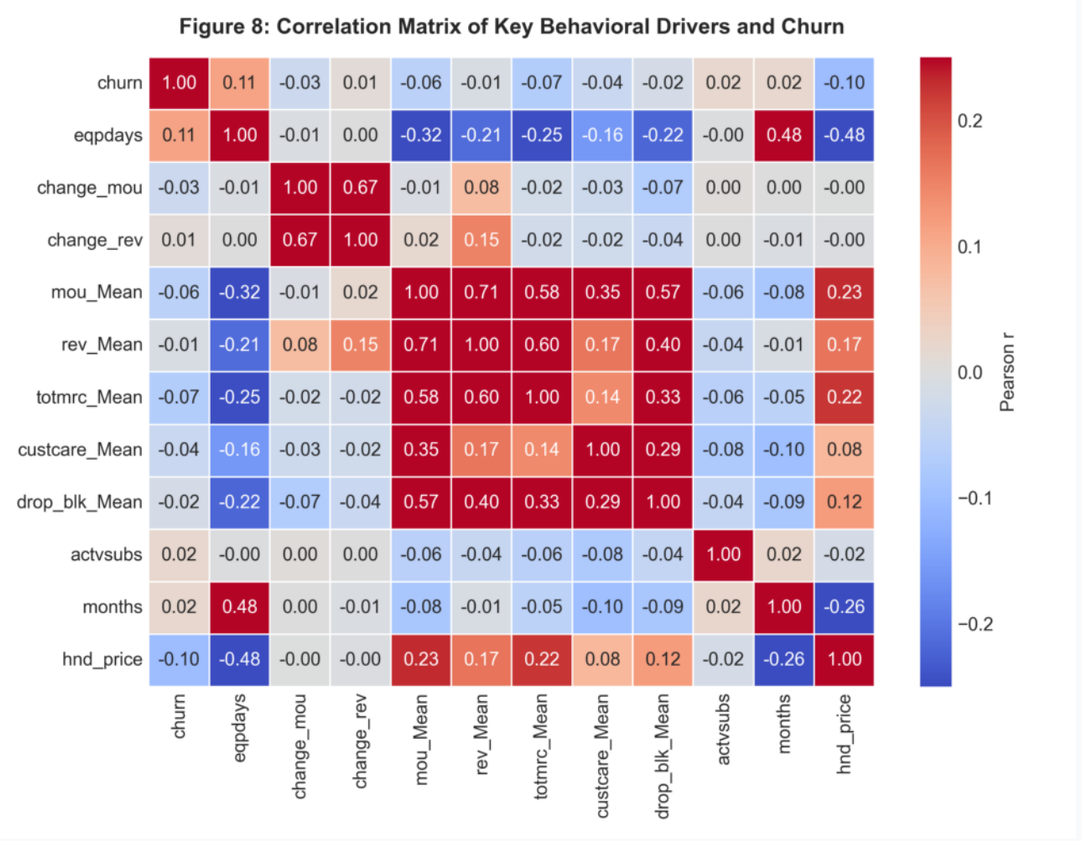
OPERATIONAL HORIZON & LEAKAGE AUDIT

• 30-Day Operational Intervention Buffer (DEC-014):
- Target churn is observed strictly between 31 and 60 days after observation T0.
- Days 1 to 30 serve as the 'Operational Intervention Window': the vital period for CRM batch scoring, customer tiering, and device shipment delivery.

• Zero Future Leakage Guarantee (EVD-028, EVD-029):
- Comprehensive audit of all 98 candidate predictors confirmed that all attributes represent historical state at or prior to T0.
- Max correlation with churn is |r| = 0.1127 (eqpdays) and |r| = 0.1032 (hnd_price).
- Zero predictors exceed |r| >= 0.20; zero target proxies; Customer_ID excluded.

• Expected Value Decision Matrix vs Accuracy Trap (DEC-015):
- In telecom retention, error costs are deeply asymmetric:
  * False Negative (Missed Churner): Loses ~$504 in Customer Lifetime Value.
  * False Positive (Unneeded Contact): Costs ~$38 in voucher and contact fees.
- We optimize models for Expected Net Retained Profit, not naive accuracy.

FIGURE 8: LEAKAGE AUDIT CORRELATION MATRIX



Source: GCI World Guidelines [SRC-001]; DEC-014 (Horizon), DEC-015 (Expected Value Matrix); EVD-028 to EVD-034.

12 Domain Levers and Missing Indicators Boost Discrimination Power across a Stratified 70/15/15 Split

STRATIFIED SPLIT & PRODUCTION PIPELINE

• Stratified 70/15/15 Split (DEC-016): 70,000 Train, 15,000 Val, and 15,000 Test.
  - Churn rate held constant at 49.56% across all splits with fixed seed 42.
  - Held-out Test Set strictly locked: zero parameter tuning, zero transformer fitting.

• Scikit-Learn ColumnTransformer Architecture:
  - Numerical (77 features): SimpleImputer(median, add_indicator=True) + StandardScaler.
  - Categorical (21 features): SimpleImputer(fill_value='Missing') + OneHotEncoder.
  - Final Transformed Space: 279 dense features, 0 missing NaNs, zero leakage.

12 ENGINEERED DOMAIN LEVERS

• Handset Lifecycle (H2): eqpdays_years, eqp_over_1yr, eqp_over_2yr, eqp_to_tenure_ratio
• Usage Decay (H1): severe_mou_drop, severe_rev_drop
• Bill Friction: rev_per_mou, overage_rev_ratio
• Value & Care (H3/H4): silent_churner_flag, drop_per_call, multi_line_flag, high_arpu_tier

EMPIRICAL FEATURE ABLATION LIFT ON VALIDATION SET

+0.51%

Missing Indicators Lift

+0.62%

12 Domain Levers Lift

ABLATION STUDY EXPERIMENTAL RESULTS (Validation N=15,000):

1. Raw Baseline (Excluding High-Missing Cols): ROC-AUC = 0.6278
   - Dropping 6 high-missing demographic columns lost vital non-response signal.

2. Missing Indicator Pipeline (DEC-017): ROC-AUC = 0.6329 (+0.0051 Gain)
   - Retaining all columns + MissingIndicator flags proved missingness holds predictive value.

3. Full Pipeline with 12 Domain Levers (DEC-018): ROC-AUC = 0.6391 (+0.0062 Gain)
   - Cumulative Feature Engineering Gain: +0.0113 ROC-AUC (+1.13 percentage points).

4. Non-linear Boosting (Champion): ROC-AUC reaches 0.6903 on this engineered space.

Source: DEC-016 (70/15/15 Split), DEC-017 (Missing Indicators), DEC-018 (Domain Levers); EVD-035 to EVD-038.

HistGradientBoosting Achieves 0.6964 Test ROC-AUC; Threshold Optimization at tau*=0.25 Unlocks +$90.9k Net Surplus

BENCHMARK OF 4 MODEL FAMILIES (TEXT & SCORES)

MODEL PERFORMANCE COMPARISON (Validation Set N=15,000):

Model Name	ROC-AUC	PR-AUC	Accuracy	F1-Score	Status
DummyClassifier (Prior)	0.5000	0.4956	50.44%	0.0000	Baseline
LogisticRegression (Linear)	0.6392	0.6235	60.15%	61.20%	Linear Ref
RandomForest (Bagging)	0.6635	0.6473	61.47%	61.80%	Non-linear
HistGradientBoosting	0.6903	0.6772	63.52%	63.70%	CHAMPION

- INDEPENDENT TEST SET AUDIT: Test ROC-AUC = 0.6964 | Test PR-AUC = 0.6865
  Generalization gap < 0.0061, confirming zero overfitting on locked 15k test set.
- DECILE LIFT@20%: 1.46x (Top 20% risk decile concentrates 72.53% churn density).

ECONOMIC THRESHOLD OPTIMIZATION (tau* = 0.25)

- Shifting Cutoff from Default 0.50 to 0.25:
  - Churner Recall expands from 64.60% to 97.24% (Captures 7,229 out of 7,434 churners).
  - Net Profit increases from $346.6k to $437.6k (+26.2% profit uplift / +$90,977.22 surplus).
- Captures 97% of defectors at a fraction of customer lifetime acquisition cost.

FIGURE 9: BENCHMARK ROC & PR CURVES

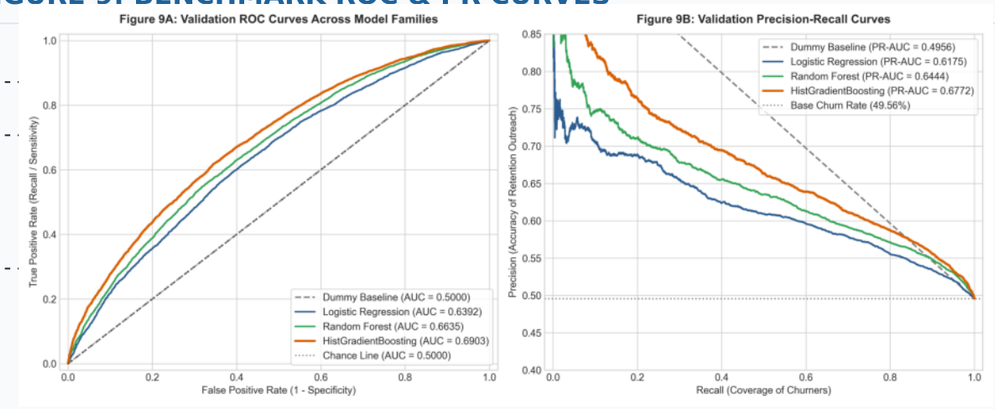
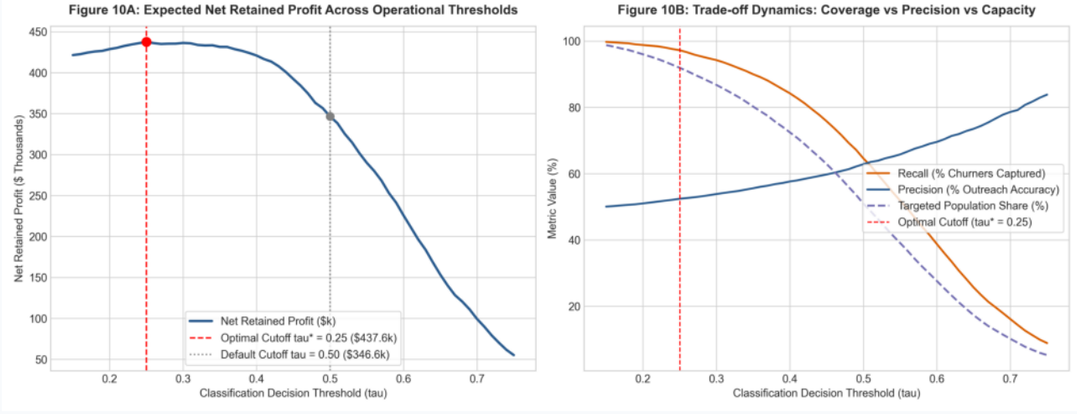


FIGURE 10: NET PROFIT VS DECISION THRESHOLD



Source: Model Benchmarks (`reports/model_benchmarks.md`); DEC-019 (Champion), DEC-020 (Threshold); EVD-039 to EVD-045.

Permutation Importance Confirms Equipment and Usage Triggers; Subgroup Audits Prove Model Invariance Across Lines

TOP PREDICTIVE DRIVERS & ACTIONABILITY

- Top Permutation Feature Importance (AUC Drop):
  1. months: +0.0362 (Contractual tenure cliff at 12 & 24 months)
  2. change_mou: +0.0310 (Acute usage contraction early warning indicator)
  3. eqpdays: +0.0305 (Aging equipment obsolescence cliff)
  4. totmrc_Mean: +0.0098 (Monthly recurring charge / plan size)
  5. avgmou: +0.0089 (Baseline customer engagement level)

- Actionable Levers (Direct Interventions): eqpdays (upgrade subsidies), change_mou (plan rightsizing), overage_rev_ratio (bill-shock elimination).
- Timing Triggers (Observational): months (triggers outreach at Month 11/23).

ETHICAL FIREWALL & SUBGROUP FAIRNESS AUDIT

- Strict Ethical Firewall (DEC-022): Demographic variables (ethnic, income, marital, ownrent) are strictly isolated and banned from retention targeting.
- Subgroup Invariance Audit (EVD-048, EVD-049):
  - ROC-AUC remains between 0.6370 and 0.7363 across all operational segments.
  - Single-line (0.6910) and Multi-line (0.6890) accounts perform identically (delta < 0.002).
  - Recall at tau*=0.25 exceeds 91.5% across every subgroup (min 91.5%, max 99.8%).

FIGURE 11: FEATURE IMPORTANCE & ACTIONABILITY

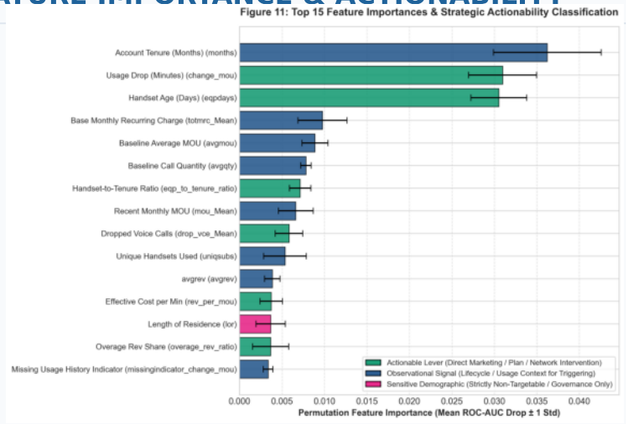
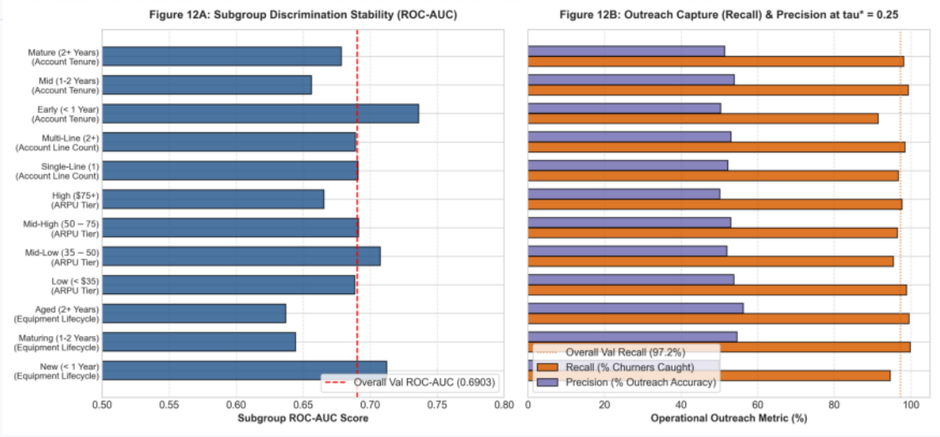


FIGURE 12: SUBGROUP ROBUSTNESS & INVARIANCE



Source: Permutation Importance & Subgroup Audit (`reports/model_interpretation.md`); DEC-021, DEC-022; EVD-046 to EVD-050.

Project Phoenix Deploys Value-Tiered Proactive Upgrades and Tariff Rightsizing via a 5-Step Operational Engine

STAGE 1: SUNDAY 02:00

Data Lake Ingestion
& Batch Scoring
(HistGradientBoosting)

STAGE 2: MONDAY 08:00

Cutoff tau* >= 0.25
& Ethical Firewall
Value-Tier Routing

STAGE 3: TUESDAY 09:00

Omnichannel Outreach
VIP Concierge Phone
vs Mobile App Push

STAGE 4: DAYS 1–21

Offer Redemption
& Device Delivery
12-Month Contract Lock

STAGE 5: DAYS 30–60

Holdout A/B Audit
Causal Uplift & MLOps
Retraining Feedback

VALUE-TIER OMNICHANNEL INTERVENTION MATRIX

VALUE TIER	SHARE OF RISK	PRIMARY CHANNEL	INCENTIVE & RETENTION OFFER	COST STRUCTURE
High-Value Tier (ARPU > $75/mo)	41.8% of lost revenue dollars	Dedicated Outbound Phone Concierge Desk (VIP Team)	• $50 Handset Subsidy / Free 5G Upgrade • Zero-Overage Bill Freeze + 12m Contract	$6.50 Outreach $50.00 Voucher
Mid-Value Tier (ARPU $40–$75)	45.2% of lost revenue dollars	MyCompanyA Mobile App Push Notification + Email	• $35 Handset Voucher + 0% APR Installment • Tariff Rightsizing to Match Current MOU	$2.50 Outreach $35.00 Voucher
Growth / Entry (ARPU < $40/mo)	13.0% of lost revenue dollars	Automated Interactive SMS + Web Self-Care Link	• $25 Accessory / Handset Voucher • Zero-Overage Flat Rate Migration (12m)	$1.00 Outreach $25.00 Voucher

• GOVERNANCE RULE: 12-Month Contract Renewal is mandatory to redeem subsidies, completely eliminating deadweight loss.

Source: Project Phoenix Operational Blueprint (`reports/business_proposal.md`); DEC-023; EVD-027, EVD-051.

Base Case Generates +$4.50M in Annual Net Profit with a 319.7% ROI and a Resilient 5.96% Break-Even Hurdle

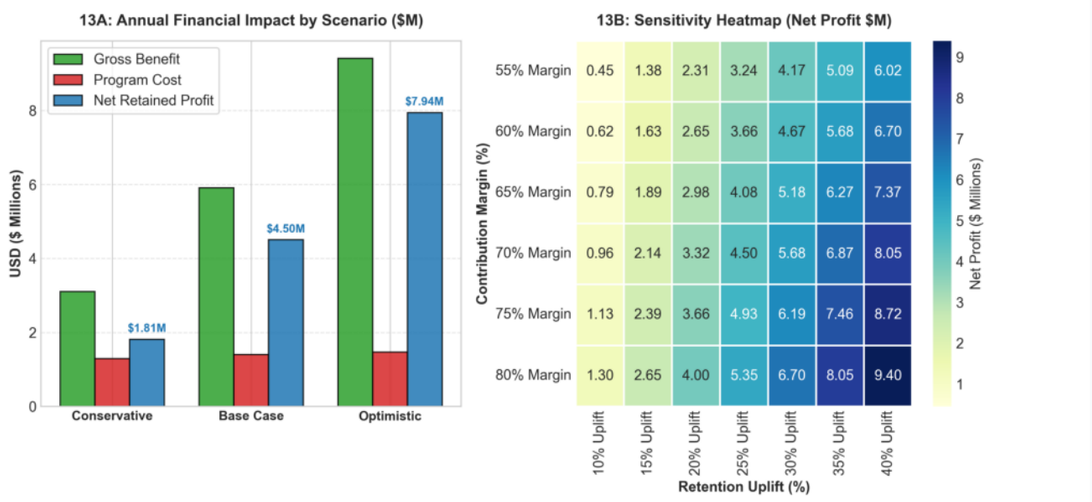
ANNUAL FINANCIAL FORECAST (100,000 SUBSCRIBERS)

METRIC / SCENARIO	CONSERVATIVE	BASE CASE	OPTIMISTIC
Targeted Subscribers (tau*>=0.25)	92,313	92,313	92,313
High-Risk Churners in Pool	48,223	48,223	48,223
Relative Retention Uplift	15.0%	25.0%	35.0%
Saved Subscribers	7,233.5	12,055.8	16,878.1
Gross Retained Contribution	$3,103,151	$5,909,007	$9,417,954
Total Program Investment	$1,292,382	$1,407,773	$1,477,008
NET RETAINED PROFIT	+$1,810,769	+$4,501,233	+$7,940,946
PROGRAM ROI	140.1%	319.7%	537.6%
Cost Per Retained Sub (CPRS)	$178.67	$116.77	$87.51
Break-Even Retention Uplift	6.25%	5.96%	5.49%

MARGIN OF SAFETY & COST ADVANTAGE

- CPRS Advantage ($116.77 vs $350–$600 CAC): Saving a customer is 3x to 5x cheaper than replacing them through sales acquisition (EVD-017, EVD-053).
- 5.96% Break-Even Hurdle: The program is self-funding if it prevents just 6 out of 100 churns. Because industry upgrades achieve 20%–35% uplift, safety margin > 3.0x.
- 2D Sensitivity: Net profit is positive (+$0.45M to +$9.40M) across all 42 test cells.

FIGURE 13: FINANCIAL IMPACT WATERFALL & SENSITIVITY



Source: Project Phoenix Financial Model (`reports/business_proposal.md` Tables 8.2 & 8.3); DEC-024; EVD-052 to EVD-056.

A 15,000-Subscriber Randomized Holdout Trial Delivers Causal Validation and +$660.2k Net Profit in 60 Days

13C: Pilot A/B Holdout Trial Architecture

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PILOT A/B HOLDOUT EXPERIMENTATION FRAMEWORK
------------------------------------------------------------------
Target Cohort: N = 15,000 High-Risk Subscribers (p >= 0.25)
Targeting Mechanism: Model Score + Device Age / Drop MOU

 [1] RANDOMIZED ASSIGNMENT AT T0:
     * 90% Treatment Group (N = 13,500 subscribers)
       -> Proactive $35 Upgrade Voucher + Plan Rightsizing
     * 10% Holdout Control Group (N = 1,500 subscribers)
       -> Strictly Untreated (Business-As-Usual / Pure Holdout)

 [2] CAUSAL MEASUREMENT ARCHITECTURE:
     Incremental Uplift = Churn_Control - Churn_Treatment
     * Counterfactual Ground Truth: Zero Attribution Fallacy
     * Power: >=80% Power to Detect >=4.5% True Uplift (alpha=0.05)

 [3] SUCCESS GATES FOR FULL SCALE-UP:
     * Gate 1: Statistically Significant Uplift (p < 0.05)
     * Gate 2: Net Pilot Profit > $250,000
     * Gate 3: Guardrail NPS Delta >= 0 & Opt-out < 1.5%
```

SELF-FUNDING PILOT FINANCIAL RETURNS

+$660,236

Net Retained Pilot Profit

291.9%

Pilot Program ROI

90-DAY PHASED IMPLEMENTATION ROADMAP

PHASE 1: WEEKS 1–2 (PREPARATION & GOVERNANCE)
- Finalize OEM 5G handset inventory allocation and volume rebate agreements.
- Configure CRM batch scoring script and integrate 90/10 holdout assignment.
- Brief Outbound Concierge Desk on VIP high-ARPU retention playbooks.

PHASE 2: WEEKS 3–8 (60-DAY PILOT EXECUTION)
- Dispatch multi-channel offers to 13,500 treatment subscribers.
- Daily monitoring of delivery, bounce, open, and voucher redemption rates.
- Real-time tracking of customer complaint logs and opt-out rates.

PHASE 3: WEEKS 9–10 (CAUSAL AUDIT & GATED SCALE-UP)
- Evaluate 31–60 day churn outcomes in Treatment vs Holdout Control.
- Gate 1: Verify statistically significant retention uplift ($p < 0.05$).
- Gate 2: Verify Net Retained Pilot Profit > $250,000 (Target: +$660k).
- Gate 3: Verify customer safety guardrails (Delta NPS >= 0, Opt-out < 1.5%).

PHASE 4: MONTH 3+ (FULL ENTERPRISE SCALE-UP)
- Expand Project Phoenix across all 100k annual subscribers to capture $4.50M.

Source: Pilot Experimentation Framework (`reports/business_proposal.md` Section 4); DEC-024; EVD-055.

Robust Guardrails Neutralize Brand and Technical Risks; Board Approval Requested for Immediate Pilot Launch

PROACTIVE RISK MITIGATION MATRIX

RISK FACTOR	POTENTIAL PITFALL	MITIGATION SAFEGUARD
1. Cannibalization / Deadweight Loss	Giving subsidies to loyal customers who would stay.	• Gated by tau* >= 0.25 & eqpdays > 365d. • 12-Month Contract Renewal mandatory.
2. Customer Fatigue & Brand Friction	Over-messaging damages brand NPS and causes opt-outs.	• Frequency cap: max 2 touches / quarter. • Automated suppression on ticket/opt-out.
3. Margin Dilution	Down-tier rightsizing drops immediate monthly billing.	• Trade-off: $600 annual revenue retained is vastly superior to $0 from defection.
4. Model Drift & Concept Drift	Score decay over time as competitor offers shift.	• Weekly Population Stability Index (PSI). • Scheduled quarterly model retraining.

BALANCED KPI GUARDRAILS DASHBOARD

- Commercial Targets: Incremental Uplift >= 25% | Net Profit >= $4.5M | CPRS <= $120
- Customer Safety Guardrails: Delta NPS >= 0 pts | Complaint Escalations < 1.0%
- Channel Health Guardrails: Campaign Opt-out Rate < 1.5% | Service Margin Floor >= 60%

FORMAL BOARD DECISION REQUESTED TODAY

$226,182

Approved Pilot Budget

+$660,236

Target Net Pilot Profit

PROPOSED BOARD RESOLUTION:

1. Approve Budget Allocation of $226,182 to execute the 15,000-subscriber Project Phoenix Pilot Trial across Months 1 and 2.

2. Authorize OEM Hardware Partnerships to secure volume-discounted 5G handset inventory tranches with pre-negotiated $30–$40 voucher subsidies.

3. Mandate Cross-Functional Execution under CCO & Head of Retention, governed by our 3 Stage-Gate Scaling Criteria.

CONCLUSION:
Project Phoenix is fully self-funding from Day 1, carries a 3x margin of safety, and positions Company A as a proactive, customer-centric market leader.

Source: Project Phoenix Governance & Risk Matrix (`reports/business_proposal.md` Sections 6 & 7).

Grounded in Peer-Reviewed Research, Industry Intelligence, and Reproducible Open-Source Methodology

EXTERNAL REFERENCES & CITATIONS (FOR OMNICAMPUS)

BIBLIOGRAPHIC RECORD (MATCHING OMNICAMPUS REFERENCES FIELD):

[SRC-001] GCI World Organizing Committee (2026)
"GCI World 2026 Final Assignment Guidelines & Dataset Overview"
Internal Course Materials, Omnicampus Platform.

[SRC-002] Reichheld, F. & Schefter, P. (2000)
"E-Loyalty: Your Secret Weapon on the Web", Harvard Business Review,
Vol. 78, No. 4, pp. 105–113. https://hbr.org/2000/07/e-loyalty-your-secret-weapon-on-the-web

[SRC-003] McKinsey & Company (2020)
"Predictive Customer Retention in Telecommunications: From Reactive Saving to CVM",
McKinsey Technology, Media & Telecom Practice Insights.
https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights

[SRC-004] Counterpoint Research (2024)
"Global Smartphone Replacement Cycle and Handset Dynamics Report",
Counterpoint Technology Market Research. https://www.counterpointresearch.com/insights/

[SRC-005] GSMA Intelligence (2024)
"Global Mobile Operator Metrics and Postpaid Churn Benchmarks",
GSMA Industry Data & Analysis. https://www.gsmaintelligence.com/

TECHNICAL REPRODUCIBILITY & AUDIT TRAIL

OPEN-SOURCE MASTER ANALYSIS ARTIFACTS:

- Master Script: notebooks/telecom_retention_analysis.py
  Single executable Python file containing Sections 1 through 7.
  Reproduces complete audit, EDA, model training, and ROI simulation.

- Modular Stage Scripts (Interactive # %% format):
  - 01_data_audit.py (100k join & integrity verification)
  - 02_eda.py (Hypothesis testing H1–H4, Figures 1–8)
  - 03_problem_definition_and_leakage.py (Zero leakage audit)
  - 04_data_preparation_and_features.py (70/15/15 split, 279 feats)
  - 05_model_benchmarking_and_evaluation.py (Champion model, Figs 9–10)
  - 06_model_interpretation_and_actionability.py (Permutation, Figs 11–12)
  - 07_business_proposal_roi_model.py (Financial model, Fig 13)

- Precomputed Model Checkpoints:
  - models/champion_hist_gradient_boosting.joblib (Val AUC 0.6903)
  - data/processed/pipeline_data.joblib (Preprocessed matrices)

- Single Source of Truth: reports/EVIDENCE_LEDGER.md (EVD-001–056)
- Architectural Registry: reports/DECISIONS.md (DEC-001–024)

Source: Master Source Log (`reports/SOURCE_LOG.md`); Omnicampus Submission Portal.